Question

Potassium salt $\mathbf{A}$ contains oxygen elements and has strong oxidizing properties. The mass fraction of potassium is 28.93%, the cation-anion ratio is 1:2, and the anion contains two elements with an atomic ratio of 1:4.

Adding pyridine and excess $\mathbf{A}$ to silver nitrate solution can produce an orange salt precipitate $\mathbf{X}$ with an oxygen mass fraction of 20.765%. Choose the option related to $\mathbf{X}$ below that has the closest numerical value and the most correct conclusion.

A. All other options are incorrect

B. In the anion and cation of $\mathbf{X}$, the coordination geometries of all metal ions are tetrahedral.

C. The cation of $\mathbf{X}$ does not obey the 18-electron rule.

D. The molar mass of the cation(s) contained in $\mathbf{X}$ is 537.3 g/mol

E. The paramagnetism of $\mathbf{X}$ is due to the presence of unpaired electrons in the contained anions.

F. The molar mass of the cation contained in $\mathbf{X}$ is 492.2 g/mol.

G. $\mathbf{X}$ The main reason for the color display is: The energy of charge transfer transitions (i.e., LMCT mechanism) from oxygen atoms to high-valent central atoms in anions is low, and the absorption of visible light displays color.

Answer

Correct Answer: C

Detailed Explanation

According to the cation-anion ratio of 1:2, first calculate the relative molar mass corresponding to 2 potassium ions as:

$$M = 2 \times 39.10 \div 28.93\% = 270.3(g/mol)$$

Then the molar mass of the anion is $270.3 - 2 \times 39.10 = 192.1(g/mol)$

Common oxidizing salts are all oxoacid salts, so it is speculated that $\mathbf{A}$ contains oxygen. Since the oxygen atom generally does not act as the central atom in the anion, assume the anion is in the form of XO_4.

Calculate the molar mass of X:

$$M(X) = 192.1 - 4 \times 16.00 = 128.1(g/mol)$$

Close to the molar mass of tellurium, but the tellurate radical has only medium oxidizing power, which does not meet the condition of strong oxidizing power, so the possibility of the anion being in the form of XO_4^{2-} is excluded.

Assume the anion is in the form of $X_2O_8^{2-}$.

Calculate the molar mass of X:

$$M(X) = (192.1 - 4 \times 16.00) \div 2 = 32.05(g/mol)$$

Close to the molar mass of sulfur. Potassium persulfate has strong oxidizing power, which meets the properties such as the cation-anion ratio of 1:2 and the atomic ratio of the two elements in the anion being 1:4. Therefore, $\mathbf{A}$ is potassium persulfate.

CHECKPOINT 2 PTS

$\mathbf{A}$ is potassium persulfate

Potassium persulfate can oxidize Ag to +2 or +3 valence. Assuming oxidation to +2 valence, the chemical formula of the precipitate can be assumed as $Ag(Py)_xS_2O_8$

Calculate the molar mass of the compound:

$$M = 8 \times 16.00 \div 20.765\% = 616.4(g/mol)$$

The number of pyridine is:

$$x = (616.4 - 107.9 - 2 \times 32.06 - 8 \times 16.00) \div 79.10 = 4.00$$

Therefore, 4 pyridine molecules coordinate to silver, which is consistent with the typical coordination number of Ag(II), so the chemical formula of the orange salt precipitate is $[Ag(C_5H_5N)_4]S_2O_8$

CHECKPOINT 2 PTS

The chemical formula of the orange salt precipitate is $[Ag(C_5H_5N)_4]S_2O_8$

The cation in the compound is $[Ag(C_5H_5N)_4]^{2+}$, which is coordinated by four pyridine molecules to Ag^{2+} in a square planar configuration through the lone pair electrons of the nitrogen atom.

Therefore, option B is incorrect.

CHECKPOINT 1 PTS

The cation $[Ag(C_5H_5N)_4]^{2+}$ of the orange salt precipitate has a square planar configuration

The molar mass of the cation:

$$M = 107.9 + 20 \times 12.01 + 20 \times 1.008 + 4 \times 14.01 = 424.3(g/mol)$$

CHECKPOINT 1 PTS

The molar mass of the cation: $M = 107.9 + 20 \times 12.01 + 20 \times 1.008 + 4 \times 14.01 = 424.3(g/mol)$

Therefore, option D is incorrect.

Since Ag(II) has a d^9 valence electron configuration, there are unpaired electrons, making the complex paramagnetic. All electrons in the anion are paired, with no paramagnetic moment. Therefore, option E is incorrect.

Check whether the cation conforms to the 18-electron rule:

$$EAN = 9 + 4 \times 2 = 17 \neq 18$$

Therefore, it does not conform to the 18-electron rule, and option C is correct.

CHECKPOINT 1 PTS

$EAN = 9 + 4 \times 2 = 17 \neq 18$ Therefore, it does not conform to the 18-electron rule, and option C is correct.

CHECKPOINT 1 PTS

Ag(II) has a d^9 valence electron configuration, so there are unpaired electrons, making the complex paramagnetic

Since the 4d orbital of Ag(II) is partially filled, the complex exhibits color through the d-d transition mechanism, which is not related to the LMCT transition that may exist in the anion, so option G is incorrect.

CHECKPOINT 1 PTS

The 4d orbital of Ag(II) is partially filled, and the complex exhibits color through the d-d transition mechanism

The molar mass of the anion:

$$M = 2 \times 32.06 + 8 \times 16.00 = 192.1(g/mol)$$

CHECKPOINT 1 PTS

The molar mass of the anion: $M = 2 \times 32.06 + 8 \times 16.00 = 192.1(g/mol)$

Therefore, option F is incorrect.

So choose option C.